2 Methodology and Experiment Setup

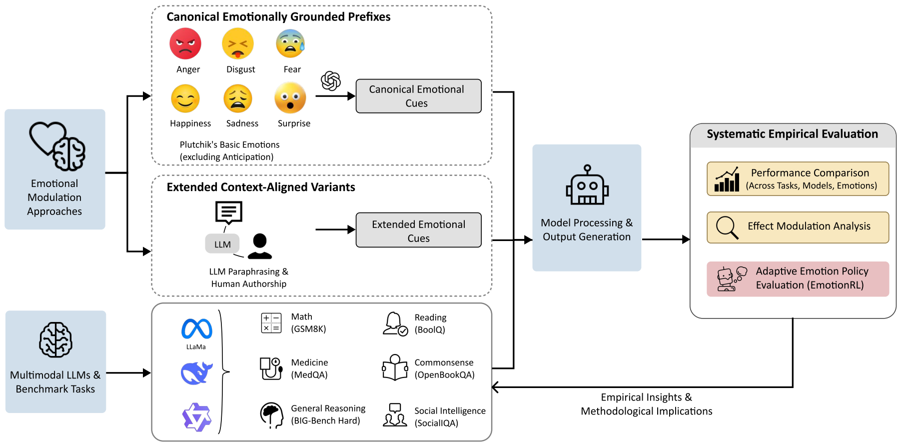

Fig. 1 Pipeline overview. We generate emotion-conditioned inputs via canonical prepended cues and context-aligned variants, evaluate them with frozen LLMs across diverse benchmarks, and analyze performance sensitivity across models, tasks, and emotions. We also include adaptive per-instance emotion selection with EmotionRL.

2.1 Design and implementation of affective manipulations

Emotion taxonomy.

Emotion conditions were operationalized using a discrete taxonomy informed by established theories of affective organization. We drew on Plutchik's basic emotions framework [33] and Russell's circumplex model of affect [34] to define six emotion categories: happiness, sadness, fear, anger, disgust, and surprise. These categories offer a compact yet diverse representation of affective states that can be succinctly instantiated in text and consistently recognized in human annotation settings. This design preserves alignment with canonical theories of emotion while ensuring that the prompt manipulation remains interpretable, controlled, and methodologically tractable.

Emotional Stimuli.

External emotional stimuli were used to manipulate the affective framing of otherwise identical questions. For each target emotion, we used GPT-4o [35] to generate short, fluent, single-sentence emotional expressions that were grammatically correct, emotionally salient, and explicitly constrained not to alter the numerical values, entities, scope, difficulty, or factual content of the underlying question. The emotional stance was framed as the speaker's attitude toward the assistant solving the problem, and could include colloquial, mildly rude, or emotionally raw language, while excluding hate speech, slurs, and explicit sexual content. For the main experiments, the GPT-4o-generated emotional sentence was prepended immediately before the original question. We adopted this design across datasets to keep the manipulation simple, interpretable, and comparable. Only for GSM8K [36], we additionally evaluated multiple insertion positions (before, within, or after the question) to test positional sensitivity.

2.2 Benchmark datasets

We evaluated model behaviour across a multi-domain benchmark spanning diverse forms of reasoning and knowledge use. GSM8K [36] measures grade-school mathematical word-problem solving with multi-step numerical reasoning. BIG-Bench Hard (BBH) [37] captures challenging general reasoning across heterogeneous tasks. MedQA (English) [38]evaluates professional-level medical question answering. BoolQ [39] tests short-passage reading comprehension with binary yes/no decisions. OpenBookQA [40] measures multiple-choice commonsense reasoning, whereas SocialIQA [41] evaluates social reasoning and everyday interpersonal inference. Collectively, these datasets cover numerical reasoning, general reasoning, domain knowledge, reading comprehension, commonsense judgment, and social inference, providing a broad behavioural evaluation suite. All datasets were used in their standard English versions, and official train, validation, and test splits were preserved whenever available to ensure comparability with prior work.

2.3 Models and inference settings

We evaluate a representative suite of both proprietary and open-weight large language models (LLMs) under strictly controlled experimental conditions. Our selection includes three state-of-the-art open-source models: Qwen3-14B [42], Llama 3.3-70B [43], and DeepSeek-V3.2 [44]. These models were chosen to encompass a diverse range of architectures, parameter scales, and training paradigms. All evaluations were conducted in a zero-shot, inference-only setting without fine-tuning, ensuring that the observed behaviors reflect the models' inherent pretrained capabilities. We utilized standardized prompt templates across all datasets to unify task descriptions and output formats. To ensure reproducibility, decoding was set to be deterministic (temperature $T = 0.0$), with all other sampling parameters, such as top-p, held at their default values.

2.4 Evaluation metrics

Our evaluation adheres to the standard task-specific protocols for each dataset. For numerical reasoning (e.g., GSM8K), a response is considered correct only if the predicted value exactly matches the ground-truth reference. For multiple-choice and binary QA tasks (e.g., MedQA-US, OpenBookQA, SocialIQA, and BoolQ), accuracy is defined as the proportion of instances where the predicted label aligns with the gold standard. These inference settings were applied consistently across all models. To isolate the impact of emotional framing, we benchmark the performance of each emotional condition against the baseline while keeping all other variables, including task instructions and decoding parameters, constant.

2.5 Validation studies and further probing

Emotion intensity study on MedQA-US

To investigate the sensitivity of model performance to the intensity of emotional framing, we conducted an exhaustive analysis using the MedQA-US dataset. For each query, we augmented the prompt with a first-person emotional prefix. Specifically, we systematically vary its intensity across three levels:

> **Slight:** "I am happy/sad/afraid/angry/disgusted/surprised."
> **Moderate:** "I am very happy/sad/afraid/angry/disgusted/surprised."
> **Extreme:** "I am extremely happy/sad/afraid/angry/disgusted/surprised."

All other components including question text, options, and task instructions remained intact to ensure controlled conditions. The original MedQA-US queries served as a neutral baseline. For consistency, we employed the same scoring protocols and metrics as in our primary experiments, enabling a direct comparative analysis across different intensity levels.

Human versus LLM emotion injection on MedQA-US.

To test whether emotional prompting effects depend on how the emotional text is constructed, we compared human-written and LLM-generated emotion prefixes on a held-out subset of MedQA-US. We randomly sampled 250 questions from the MedQA-US pool used in the main experiment. Human-written prefixes were produced by four volunteer graduate-student annotators recruited from multiple institutions. For each question, annotators wrote first-person emotional sentences for six target emotions (*happiness, anger, fear, sadness, surprise*, and *disgust*). All annotations were completed without AI assistance, and annotators were instructed to ensure that each sentence was emotionally explicit, contextually appropriate, and suitable for direct prepending to the original question. We then evaluated the same question set under two parallel conditions: one using LLM-generated emotional prefixes, as in the main prepended-emotion setting, and the other using the human-written prefixes. All evaluations were conducted with Qwen3-14B under identical decoding settings. Of the 250 sampled questions, 219 had usable gold-answer labels for automatic scoring; the remaining items were excluded because no valid reference answer could be consistently recovered from the sampled subset. Accuracy was computed separately for the neutral baseline and for each of the six emotion conditions in both the human-injected and LLM-injected settings.

2.6 EmotionRL

Problem definition

We formulate adaptive emotional prompting as an instance-wise decision problem over a fixed emotion set $\mathcal{A} = \{\texttt{ANGER}, \texttt{DISGUST}, \texttt{FEAR}, \texttt{HAPPINESS}, \texttt{SADNESS}, \texttt{SURPRISE}\}$. For each input question x_i, the agent selects one emotion $a \in \mathcal{A}$, prepends the corresponding emotional expression to the original prompt, and submits the resulting prompt to a frozen backbone LLM, which then produces the final answer.